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AMR & AGV: The Power Behind the Industrial 4.0 Revolution

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Battery Solutions for AMR, AGV, UGV & Mobile Robotics | Custom Li-ion Battery Packs | OEM/ODM | BMS & CAN...

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An AMR can be autonomous.

It can navigate around people, optimize its route and communicate with a fleet-management system.

But there is still one very basic question:

How long can it keep moving?

As AMR and AGV fleets become more automated, battery performance is becoming closely connected to uptime, charging strategy and productivity.

That is why the battery deserves more attention than simply asking:

"How many Ah?"

From Automated Vehicles to Autonomous Mobility

AGVs have been used in factories and warehouses for decades, mainly following predefined routes and operating within relatively structured environments.

AMRs have taken industrial mobility a step further.

With technologies such as LiDAR, machine vision, SLAM and fleet-management software, AMRs can navigate dynamically, avoid obstacles and adapt to changing working conditions.

This flexibility is one of the reasons AMRs are becoming increasingly attractive for modern manufacturing and intralogistics.

But greater autonomy also changes the role of the battery.

The robot is no longer simply moving from Point A to Point B.

It is continuously making decisions about:

- Where to go
- What task to perform
- When to return for charging
- Whether it has enough energy for the next mission

This means the battery has become part of the robot's overall operating system.

Battery → BMS → Robot → Charger → Fleet Management

And when these systems are properly integrated, energy management becomes much more than a charging problem.

It becomes a productivity issue.

A Real-World Example: Charging Can Affect Productivity

One example that illustrates this well comes from **Mauser Packaging Solutions**.

According to OTTO Motors' published case study, the company's previous AGV system typically completed around **3–4 missions before returning to the charging station**.

After deploying OTTO 1500 AMRs, the vehicles could complete **up to 36 missions before returning to charge**, using opportunity charging between jobs.

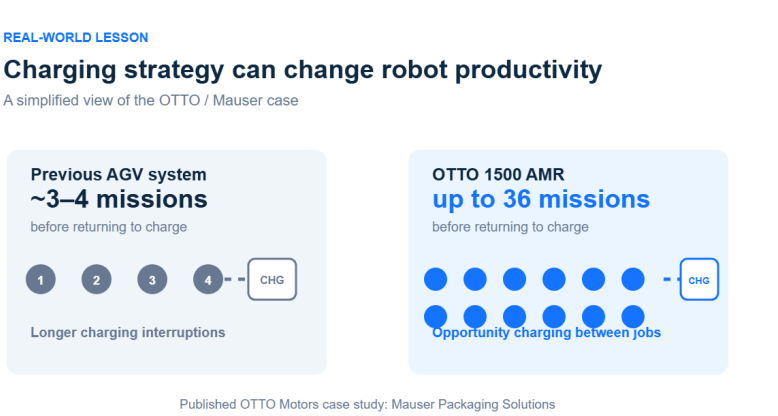
OTTO reported a significant increase in throughput at the facility.

The interesting part is not simply the difference between two robot platforms.

It is the role that **charging strategy** played in the operation.

In a high-utilization environment, a robot sitting at a charging station is not moving materials.

If charging can happen during natural gaps between missions, the battery can support much higher vehicle availability without simply making the battery larger.



That is an important shift in the way we should think about AMR battery design.

Why Battery Capacity Is Only Part of the Equation

When an AMR manufacturer starts discussing a new battery, the conversation often begins with voltage and capacity:

> 48V? 100Ah? 200Ah?

These numbers are important.

But they don't tell the whole story.

Two batteries with the same nominal voltage and capacity can behave very differently in a real AMR because of differences in cell chemistry, discharge capability, BMS configuration, thermal conditions and charging strategy.

For an industrial robot, the more useful question is:

What does the battery need to do during the actual duty cycle?

What Actually Matters in an AMR/AGV Battery?

Battery Factor	Why It Matters
Voltage & Capacity	Available energy and system compatibility
Continuous / Peak Current	Acceleration, payload and demanding operating conditions
Charging Current	How quickly the robot can return to service
Cycle Life	Important for high-utilization fleets
Temperature Range	Directly affects performance and battery life
BMS Communication	Enables the robot to monitor battery status
SOC Accuracy	Helps fleet software make better charging decisions
Mechanical Design	Space, weight, mounting and connector requirements

For example, a robot that operates on a flat warehouse floor with a relatively stable payload has a very different energy profile from one that frequently accelerates, climbs ramps or carries heavy loads.

The battery should therefore be designed around the **real operating profile**, not just the nominal capacity.

The BMS Is Becoming Part of the Robot

Another area that deserves more attention is the BMS.

In a basic battery application, the BMS may primarily be viewed as a protection device.

For an AMR or AGV, its role can be much broader.

The robot may need access to information such as:

- State of Charge (SOC)
- State of Health (SOH)
- Pack voltage
- Current
- Cell voltage
- Temperature
- Charging status
- Fault and protection information

Depending on the architecture, communication may use **CAN, RS485, UART or other protocols**.

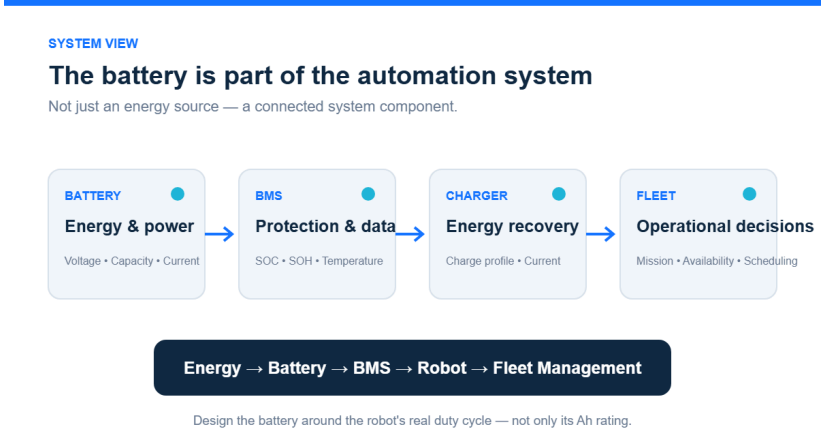
Why does this matter?

Imagine a fleet of 50 AMRs.

If the fleet-management system only knows that a robot has "low battery", its options are limited.

But if it knows the battery's SOC, health condition, temperature and charging status, it can make much better decisions about which robot should take the next mission and which one should charge.

This is where the battery starts becoming part of the **automation data loop**, rather than simply an energy source.



Opportunity Charging Changes the Design Question

Traditional battery thinking often looks like this:

More runtime = larger battery

But this is not always the best answer for an AMR.

A larger battery also means:

- More weight
- More installation space
- Higher cost
- Potentially longer charging requirements

Opportunity charging creates another possibility.

Instead of waiting until the battery is nearly empty, the robot can use available time between missions to recharge.

This changes the design objective from:

"How large does the battery need to be?"

to:

"How efficiently can the robot use the energy available to it?"

That difference can have a significant impact on the overall system design.

What AMR & AGV Manufacturers Should Define Before Choosing a Battery

Battery selection becomes much easier when the application requirements are clearly defined at the beginning.

Before Choosing a Battery, Define These Parameters:

Area	What to Define
Operating Profile	Hours per shift / missions per day
Payload	Typical and maximum payload
Current Demand	Continuous and peak current
Charging	Standard, fast or opportunity charging
Environment	Temperature, humidity and indoor/outdoor use
Communication	CAN, RS485, UART or other protocol
Mechanical Integration	Dimensions, weight, connectors and mounting
Service Life	Expected cycles and replacement strategy

These details allow the battery supplier to work backwards from the actual application.

Instead of starting with:

> "We need a 48V 100Ah battery."

the engineering discussion becomes:

> "This robot carries X kg, operates X hours per day, has a peak current of X A, uses opportunity charging and needs CAN communication."

That is a much better starting point for battery design.

The Real Cost of a Battery Is Not Just Its Purchase Price

For an AMR manufacturer, comparing batteries purely on purchase price can be misleading.

A lower-cost battery may look attractive initially.

But what happens if it has:

- Shorter cycle life?
- Lower usable capacity?
- Longer charging time?
- Poor SOC accuracy?
- Limited communication capability?
- More frequent replacement?
- Higher integration costs?

The battery's real cost should be considered together with **downtime, replacement frequency, charging infrastructure and engineering integration**.

In other words:

The cheapest battery is not necessarily the lowest-cost battery.

For a fleet operator, what matters most is often much simpler:

How much time is the robot actually available for work?

The Goal Is Not the Biggest Battery

As AMRs and AGVs become more capable, battery design will become increasingly integrated with the rest of the vehicle.

Cell selection matters.

BMS matters.

Charging matters.

Communication matters.

Mechanical integration matters.

But none of these should be considered independently.

The best solution is the one that balances all of them around the robot's actual operating requirements.

Sometimes that means a larger battery.

Sometimes it means faster charging.

Sometimes it means better energy management.

And sometimes it means a smarter BMS.

The goal is not to build the biggest battery.

The goal is to keep the robot working.

Mine Thought

Industry 4.0 is ultimately about making industrial systems more connected, more flexible and more efficient.

AMRs and AGVs are a clear example of this transformation.

But autonomy does not eliminate the importance of the power system.

It makes it more important.

A modern AMR battery should not be viewed simply as a box containing cells.

It is part of a larger system involving the **robot, BMS, charger and fleet-management platform**.

And the earlier this is considered in the development process, the easier it becomes to optimize the final system.

The future of industrial automation will be increasingly autonomous.

Its power systems need to be designed with the same level of intelligence.

For AMR and AGV manufacturers, battery development is rarely just about voltage and capacity. The real challenge is finding the right balance between energy, power, charging, communication, mechanical integration and long-term reliability.

#AMR #AGV #MobileRobotics #IndustrialAutomation #Robotics #BatteryTechnology #LithiumIonBattery

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